

Architectural Paradigms for Global Personal Identity: A Comprehensive Analysis of Naming Conventions and Information Systems

The design of an information architecture capable of representing personal names on a global scale requires a departure from the reductionist models prevalent in Western software engineering. While the simplified bipartite structure of "given name" and "surname" serves as a convenient shorthand for Euro-centric data entry, it fails to account for the ontological diversity of identity markers across the world's cultures. The strategy of representing names as a single string accompanied by a "type" metadata tag—such as legal, former, or preferred—is a pragmatic starting point that avoids immediate structural failures, but it introduces significant downstream challenges for programmatic distinguishability, entity resolution, and cultural respect.¹ This report evaluates the current state of naming convention research, identifies critical gaps in existing architectural thinking, and proposes a framework for a more robust, internationally aware identity system.

The Ontological Complexity of Global Naming

The foundational challenge in name representation is the lack of a universal definition for what constitutes a "name part." The Western assumption that a name consists of a personal identifier (given name) and a lineage identifier (family name) is routinely contradicted by global practice. In Iceland, for example, the concept of a fixed family name is virtually non-existent for the majority of the population. Instead, the Icelandic naming system utilizes patronymics or matronymics, where the secondary identifier is a description of the father's or mother's name followed by a gendered suffix—"son" for males and "-dóttir" for females.¹ In this context, a father and daughter do not share a surname, rendering familial grouping by string comparison impossible. Furthermore, Icelandic telephone directories are sorted by given name, highlighting that even the "canonical" display order is locally defined and contextually dependent.¹

This complexity extends to the number of name components. While the Western template allows for a middle name, many cultures utilize single names or extensive chains of identifiers. Mononymous names are standard in parts of Indonesia, Burma, and Southern India.¹ Conversely, Spanish and Portuguese naming traditions involve multiple family names. A Spanish individual typically carries two surnames: the first from the father's first surname and the second from the mother's first surname.⁵ Portuguese-speaking cultures, such as those in Brazil, often reverse this order, placing the maternal surname first.¹ These family names are often linked by particles such as "de" or "e," which are integral to the name and should not be

discarded during normalization.⁷

Comparative Framework of Global Naming Conventions

Cultural Context	Primary Name Order	Functional Components	Address Norms
Ibero-American	Given-Paternal-Maternal	Two family names; maternal name often dropped informally.	Formal: Mr.. ⁶
Arabic	Given-Patronymic-Nisbah	Genealogical chain (son of, father of); honorifics common.	Formal: Use of Kunya (Abu). ⁹
Icelandic	Given-Patronymic	No fixed family name; last name describes parentage.	Formal: Use of Given Name only. ³
Sinitic (Chinese)	Family-Given	Family name is first; given name often two syllables.	Formal: Mr. [Family Name]. ¹
Vietnamese	Family-Middle-Given	Three main components; given name is primary address.	Formal: Mr. [Given Name]. ¹⁰
Hungarian	Family-Given	Only European culture placing family name first.	Formal: Mr. [Family Name]. ¹¹

Gaps in the Single-String Architectural Approach

The proposed approach of utilizing a single string with a "type" metadata tag effectively circumvents the immediate structural errors associated with forced field splitting. However, this model suffers from several deep-seated architectural gaps that become apparent during third-party integration and complex data processing.

One primary gap involves the loss of semantic intent required for sorting and collation. Information systems often require names to be ordered alphabetically by their most "significant" part—typically the family name in the West. Without structured fields, a system cannot differentiate between "Mao Ze Dong" (where the first word is the family name) and "John Smith" (where the last word is the family name) without resorting to sophisticated, and often error-prone, natural language processing.⁷ This lack of distinguishability forces the architecture to perform "sense-making" repeatedly at the application layer, rather than having a reliable data model to draw upon.¹³

A second gap is found in the management of transliteration and script variations. Representing a name as a single string assumes that the script used is sufficient for all system functions. However, languages like Japanese utilize multiple scripts—Kanji for meaning, Kana for phonetic pronunciation, and Romaji for Latin transcription.¹ Because Kanji characters can have multiple readings, storing only the name string without a separate phonetic representation (furigana) makes it impossible for software to reliably sort Japanese names or for users to know how to pronounce them.¹

Technical Implications of Name Component Misidentification

Use Case	Requirement	Single-String Limitation
Formal Mailing	Mr./Ms. [Family Name]	Unable to extract family name across cultures. ⁷
Alphabetical Sorting	Order by [Primary Identifier]	Inversion of order in East Asia/Hungary causes mis-sort. ¹
Deduplication	Exact or fuzzy matching	Middle names/initials treated inconsistently. ¹³
Legal Compliance	Match Passport/ID	Transliteration in ICAO zone differs from display. ¹⁴
Pronunciation	Phonetic string (e.g., Kana)	Original script may have ambiguous readings. ¹

Technical Standards for Identity Representation

To move beyond the limitations of the single-string model, architectures should look toward established international standards that provide more granular, yet flexible, structures for representing identity.

The vCard 4.0 Standard (RFC 6350)

The vCard 4.0 specification provides a robust framework for name representation that balances simplicity with international needs. It differentiates between the "Formatted Name" (FN), which is the full string intended for display, and the "Structured Name" (N), which breaks the identity into five components: Family Name, Given Name, Additional Name(s), Honorific Prefix(es), and Honorific Suffix(es).¹⁵ Crucially, vCard 4.0 mandates the use of the UTF-8 character set, eliminating the historic issues associated with disparate encodings.¹⁶

Furthermore, the standard introduces the SORT-AS parameter, which allows an architect to store a separate string specifically for sorting purposes. This is an essential edge case for names containing particles like "van" or "von," which may be ignored in sorting in some cultures (like the Netherlands) but included in others (like Belgium).¹⁷ It also accounts for the Hungarian practice of placing surnames first while ensuring they are still correctly categorized as family names in a global database.¹

ICAO Document 9303 and Machine-Readable Travel Documents

The International Civil Aviation Organization (ICAO) defines the standards for Machine-Readable Travel Documents (MRTDs), which are the global "source of truth" for legal names. The Machine-Readable Zone (MRZ) on a passport is a highly constrained environment, using only the Latin characters A-Z, numbers 0-9, and the filler character <.¹⁴ The MRZ structure typically separates names into a "Primary Identifier" (position 6 onwards) and a "Secondary Identifier".¹⁸

A significant gap in architectural thinking is the failure to realize that the name on the visual page of a passport may differ significantly from the name in the MRZ. Diacritics are removed, hyphens are replaced by filler arrows, and apostrophes are deleted entirely.¹⁸ Architects must decide whether their system should store the "clean" visual name, the "standardized" MRZ name, or both. For systems involving travel, banking, or legal verification, matching the MRZ exactly is often a functional requirement that a single-string display name cannot satisfy.¹⁴

Comparison of Name Formatting Standards

Standard	Primary Identifier	Secondary Identifiers	Support for Non-Latin Scripts
vCard 4.0	Family Name	Given, Additional, Prefix, Suffix	Full (UTF-8, LANGUAGE param) ¹⁶
ICAO 9303	Primary Identifier	Secondary Identifier	None (Transliterated to

			Latin A-Z) ¹⁴
Google People API	familyName	givenName, middleName	Full (Phonetic fields included) ²⁰
LinkedIn API	lastName	firstName, maidenName	Multi-locale support for translations. ²¹

The Challenge of Name Parsing and Algorithmic "Sense-Making"

As the proposed architecture attempts to match name forms from third-party sources, it will inevitably encounter the limitations of automated parsing. Name parsing is the process of breaking a raw string into semantic elements such as genus, authorship, or in the case of humans, given and family names.²² Traditionally, this was handled via regular expressions or rule-based systems, but these are notoriously brittle when confronted with the ambiguity of global names.²²

Consider the three-word name "David Lloyd George." A rule-based parser cannot determine if this represents two given names and a single family name ("David Lloyd" + "George") or a given name followed by a double-barreled family name ("David" + "Lloyd George").⁷ This ambiguity is a frequent source of error in identity resolution systems. Modern approaches are shifting toward probabilistic models and Large Language Models (LLMs) to perform name extraction.²⁴ LLMs can be prompted with cultural context—such as the country of origin—to more accurately distinguish between name parts, although they require careful validation to ensure they do not "hallucinate" or drift from the required schema.²⁵

Performance and Reliability of Parsing Methodologies

Method	Accuracy on Clean Data	Handling of Multilingual Noise	Computational Cost
Regular Expressions	High (for known patterns)	Very Low	Minimal ²²
Rule-Based (e.g., nameparser)	Medium	Low (English-centric)	Low ²³
Probabilistic (e.g.,	High	Medium	Moderate ²⁴

HMM)			
Generative AI (LLM)	Very High	Very High	High ²⁷

Edge Cases and Unaware Concerns in Information Architecture

In addition to the core cultural variations, there are several technical and social edge cases that should influence the design of a personal name architecture.

Mononymous Name Workarounds

The requirement for both a "First Name" and "Last Name" in many legacy systems (such as PeopleSoft) creates significant issues for mononymous users. Standardizing the placeholder for the "empty" field is essential for data integrity. While some systems use "FNU" (First Name Unknown), this is factually incorrect for someone who *has* a name but lacks a surname. The Data Governance Committee (DGC) of the SBCTC determined that the best practice for such systems is to place the legal name in the "Last Name" field and utilize a single hyphen ("-") as a placeholder in the "First Name" field, as it is visible and lacks the negative connotations of "FNU" or "N/A".⁴

The Lifecycle of Identity and Name Changes

A static name model fails to account for the fact that names are dynamic. Beyond marriage and divorce, name changes occur due to gender transition, religious conversion, or even the desire to avert bad luck, as is common in Thailand.⁸ For many individuals, such as transgender people or victims of abuse, the "former" name is a "deadname" that should be treated with extreme privacy or removed entirely from display to avoid psychological harm or safety risks.²⁸ The architecture must distinguish not just the *type* of name but also its *visibility* and *context* (e.g., "Legal only," "Internal only," "Public").

Casing and Diacritical Sensitivity

Automatic casing normalization—such as forcing a string to Title Case—is a common pitfall. Names like "McNamara," "van der Waals," or the Irish "ffrench" rely on specific capitalization that carries semantic meaning.¹ Similarly, the omission of diacritics in Vietnamese names can change the meaning of the name entirely; names with the same spelling but different tones represent different characters.¹⁰ A robust architecture must preserve the user's input exactly as entered (using Unicode) and avoid destructive normalization.¹

Toward a Robust Identity Framework: Best Practices

The transition from a single-string model to a semantically aware identity system requires a

multi-layered approach. The following recommendations represent the established best practices for global name architecture.

1. Separation of Display and Semantic Fields

The architecture should maintain a "Full Name" string for display purposes, allowing the user to dictate the exact presentation order and punctuation of their name. However, this should be supplemented by a structured object (potentially stored as JSON in a relational database or as a separate table) that identifies the given name(s), family name(s), and primary identifier for sorting.¹³

2. Implementation of Locale-Aware Collation

Sorting should not be a simple ASCII-based operation. Instead, the architecture should leverage the Unicode Collation Algorithm (UCA) and specify the locale of the data. This ensures that names containing "ch" in Spanish or "Å" in Danish are sorted according to the expectations of those cultures.³²

3. Support for Multiple Scripts and Phonetics

For data originating in East Asia, the architecture must support multiple script representations for a single identity. Storing the "Reading" or "Phonetic" version of a name is not an edge case but a requirement for accurate sorting and search in Japan and China.¹

4. Inclusive Metadata and Privacy Controls

Names should be accompanied by metadata that goes beyond simple "types." Fields should indicate the language of the name, the script used, and the privacy level of that specific name version.²⁰ This allows for the sensitive handling of preferred and legal names in diverse professional and personal contexts.

Detailed Analysis of Global Sorting and Collation Requirements

When names are stored as a single string, the system loses the ability to perform locale-sensitive sorting, which is a critical requirement for any global-scale application. The following table illustrates the significant variations in primary and secondary sorting levels across different locales.

Locale	Character(s)	Sorting Rule / Behavior	Impact on Search/Sorting
Spanish (Trad.)	ch, ll	Treated as distinct letters between c/d	Mis-sorted if treated as individual

		and l/m.	characters. ³²
German (Phone)	ä, ö, ü	Sorted as if they were ae, oe, ue.	"Müller" comes before "Mustermann." ³²
Danish/Danish	Å, aa	Sorted at the end of the alphabet, after Z.	"Åberg" is at the very end of the list. ³²
Swedish	v, w	Traditionally treated as variants of the same letter.	"Wickström" sorts with "Vickström." ³²
French	Accents	Sorted in backwards order from the end of the string.	"côte" sorts before "coté." ³²
Icelandic	All	Sorted by Given Name, not by Patronymic.	Fundamental shift in UI design for directories. ¹

The Role of the Unicode Collation Algorithm (UCA)

The UCA provides a multi-level comparison key system that allows architects to define how granular their sorting should be. A primary level difference (e.g., a vs b) is the most significant, followed by secondary (accents) and tertiary (case) differences.³³ By leveraging UCA-based collations (such as those provided in IBM DB2 or SQL Server), an architect can ensure that search results and lists are culturally appropriate without manually implementing complex sorting logic for every language.³⁶

Architectural Synthesis: The Hybrid Identity Model

The optimal architecture for representing names in a global context is a hybrid model that preserves the flexibility of the user's intent while providing the structure necessary for programmatic reliability.

The Canonical "Display" String

The "Display Name" or "Full Name" should be a single, unrestricted UTF-8 string. This is the version the user provides as their primary identity and should be used for all informal greetings (e.g., "Welcome back,").⁷ This field avoids the "first/last" trap and naturally supports

mononymous names, specific punctuation, and non-Western name orders.

The Semantic "Metadata" Object

Attached to the display string should be a structured metadata object. This object does not dictate the display but provides the "hints" necessary for the system to make sense of the data.

- **Primary Identifier:** The token(s) used for formal address and primary sorting.
- **Secondary Identifiers:** Additional tokens (middle names, generational qualifiers).
- **Phonetic Representation:** Essential for Sinitic scripts and Japanese.
- **Script and Locale:** The linguistic context of the name.
- **Legal vs. Social Flags:** Indication of which name is used for which context.

Example Schema Mapping for Global Names

Person	Display Name	Primary ID	Secondary IDs	Locale
A	Mao Ze Dong	Mao	Ze Dong	zh-CN ¹
B	Björk Guðmundsdóttir	Björk	Guðmundsdóttir	is-IS ¹
C	María José García López	García López	María José	es-ES ⁶
D	Cher	Cher	(none)	en-US ⁴
E	Prince	Prince	(none)	en-US ¹³
F	van der Meer, Karen	van der Meer	Karen	nl-NL ³⁸

Conclusion: Designing for the Human Element of Data

Information architecture is often treated as a purely technical exercise, yet in the domain of personal names, it is a socio-technical one. Names are the primary markers of identity, heritage, and respect. An architecture that represents names solely as strings with simple types avoids complexity but at the cost of functionality and inclusivity. By incorporating established standards like vCard 4.0, understanding the constraints of legal identity systems like ICAO 9303, and leveraging the semantic power of locale-aware collation and phonetic storage, architects can

build systems that are truly global in scope.

The primary gap in the current thinking is the assumption that programmatic distinguishability is a secondary concern. In reality, as systems scale and integrate with third-party data providers, the ability to semantically "understand" a name becomes a functional requirement for security, compliance, and user experience. Moving toward a hybrid model—where a single display string is enriched by semantically typed metadata—is the most resilient path forward for modern identity management systems. This approach ensures that whether a user is mononymous, utilizes multiple family names, or follows a patronymic tradition, their identity is accurately captured and respectfully utilized across the digital landscape.

Works cited

1. Personal names around the world - W3C, accessed May 3, 2026, <https://www.w3.org/International/questions/qa-personal-names>
2. When Names Don't Fit the Form: How Western Identity Systems Are Failing the Rest of the World | by Paul Wong | Medium, accessed May 3, 2026, <https://medium.com/@plwong96uk/when-names-dont-fit-the-form-how-western-identity-systems-are-failing-the-rest-of-the-world-3ca34a5041e4>
3. Icelandic Naming System: A Unique Tradition in Modern Times, accessed May 3, 2026, <https://www.icls.edu/blog/icelandic-naming-system>
4. Mononyms (or single names) in PeopleSoft, accessed May 3, 2026, <https://www.sbctc.edu/resources/documents/colleges-staff/commissions-councils/dgc/dgc-mononyms-decision.pdf>
5. Dos Apellidos: When Families Have Two Surnames | Denver Public Library Special Collections and Archives, accessed May 3, 2026, <https://history.denverlibrary.org/news/genealogy/dos-apellidos-when-families-have-two-surnames>
6. Spanish naming customs - Wikipedia, accessed May 3, 2026, https://en.wikipedia.org/wiki/Spanish_naming_customs
7. Personal names - Internationalization, accessed May 3, 2026, https://www.w3.org/International/wiki/Personal_names
8. Falsehoods Programmers Believe About Names – With Examples - Shine Solutions Group, accessed May 3, 2026, <https://shinesolutions.com/2018/01/08/falsehoods-programmers-believe-about-names-with-examples/>
9. The Typology of Arabic Proper Nouns | PDF - Scribd, accessed May 3, 2026, <https://www.scribd.com/document/399051479/The-Typology-of-Arabic-Proper-Nouns>
10. Vietnamese name - Wikipedia, accessed May 3, 2026, https://en.wikipedia.org/wiki/Vietnamese_name
11. UsefulNotes / Name Order Conventions - TV Tropes, accessed May 3, 2026, <https://tvtropes.org/pmwiki/pmwiki.php/UsefulNotes/NameOrderConventions>
12. Chinese transliteration overview - IBM, accessed May 3, 2026, <https://www.ibm.com/docs/en/ignm/7.0.0?topic=transliteration-chinese-overview>

13. Proper Storage and Handling of Personal Names - SQLServerCentral, accessed May 3, 2026, <https://www.sqlservercentral.com/articles/proper-storage-and-handling-of-personal-names>
14. Machine-readable passport - Wikipedia, accessed May 3, 2026, https://en.wikipedia.org/wiki/Machine-readable_passport
15. vCard - Wikipedia, accessed May 3, 2026, <https://en.wikipedia.org/wiki/VCard>
16. vCard 4.0 - Introduction - CalConnect, accessed May 3, 2026, <https://devguide.calconnect.org/vCard/vcard-4/>
17. Wikipedia: Categorization/Sorting names, accessed May 3, 2026, https://en.wikipedia.org/wiki/Wikipedia: Categorization/Sorting_names
18. STUDENT NAME GUIDELINE, accessed May 3, 2026, <https://policies.newcastle.edu.au/download.php?id=183&version=4&associated>
19. Falsehoods programmers believe about names : r/programming - Reddit, accessed May 3, 2026, https://www.reddit.com/r/programming/comments/191ot55/falsehoods_programmers_believe_about_names/
20. REST Resource: people | People API - Google for Developers, accessed May 3, 2026, <https://developers.google.com/people/api/rest/v1/people>
21. Basic Profile Fields - LinkedIn | Microsoft Learn, accessed May 3, 2026, <https://learn.microsoft.com/en-us/linkedin/shared/references/v2/profile/basic-profile>
22. "gnparser": a powerful parser for scientific names based on Parsing Expression Grammar - PMC, accessed May 3, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC5446698/>
23. GitHub - derek73/python-nameparser: A simple Python module for parsing human names into their individual components, accessed May 3, 2026, <https://github.com/derek73/python-nameparser>
24. [2601.18014] A System for Name and Address Parsing with Large Language Models - arXiv, accessed May 3, 2026, <https://arxiv.org/abs/2601.18014>
25. A System for Name and Address Parsing with Large Language Models - arXiv, accessed May 3, 2026, <https://arxiv.org/html/2601.18014v1>
26. LLM Output Parsing and Structured Generation Guide - Tetrade, accessed May 3, 2026, <https://tetrade.io/learn/ai/llm-output-parsing-structured-generation>
27. A System for Name and Address Parsing with Large Language Models - ResearchGate, accessed May 3, 2026, https://www.researchgate.net/publication/400084181_A_System_for_Name_and_Address_Parsing_with_Large_Language_Models
28. Use of chosen names in electronic health records to promote transgender inclusivity - PMC, accessed May 3, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10198525/>
29. Talking tech: Handling human names - IAPP, accessed May 3, 2026, <https://iapp.org/news/a/talking-tech-handling-human-names>
30. DB Design Question - Storing International People Names - Stack Overflow, accessed May 3, 2026,

<https://stackoverflow.com/questions/4611967/db-design-question-storing-international-people-names>

31. How to organize multilingual fields in the database ? : r/Nestjs_framework - Reddit, accessed May 3, 2026, https://www.reddit.com/r/Nestjs_framework/comments/1i7ljgr/how_to_organize_multilingual_fields_in_the/
32. Collation | ICU Documentation, accessed May 3, 2026, <https://unicode-org.github.io/icu/userguide/collation/>
33. Character data, Unicode and Collations - Mimer Developer, accessed May 3, 2026, <https://developer.mimer.com/article/character-data-unicode-and-collations/>
34. About internationalization - I18n - W3C, accessed May 3, 2026, <https://www.w3.org/International/i18n-drafts/nav/about>
35. What to Include In Your LinkedIn Name field!! | by Dennis Olympios | Medium, accessed May 3, 2026, <https://dennisolympios.medium.com/what-to-include-in-your-linkedin-name-field-ec95521137f>
36. icu/docs/userguide/collation/customization/index.md at main · unicode-org/icu - GitHub, accessed May 3, 2026, <https://github.com/unicode-org/icu/blob/main/docs/userguide/collation/customization/index.md>
37. Unicode Collation Algorithm based collations - IBM, accessed May 3, 2026, <https://www.ibm.com/docs/en/db2/11.5.x?topic=collation-unicode-algorithm-based-collations>
38. Parsed names - IBM, accessed May 3, 2026, https://www.ibm.com/docs/SSEV5M_5.0.0/com.ibm.iis.gnm.names.doc/topics/gnr_gnm_con_parsednames.html